

Three Derived Populations from the Cleaned Penetration Archive Structure, Role, and Thesis Update Plan

Mie Spray Postprocessing Project

2026 年 5 月 22 日

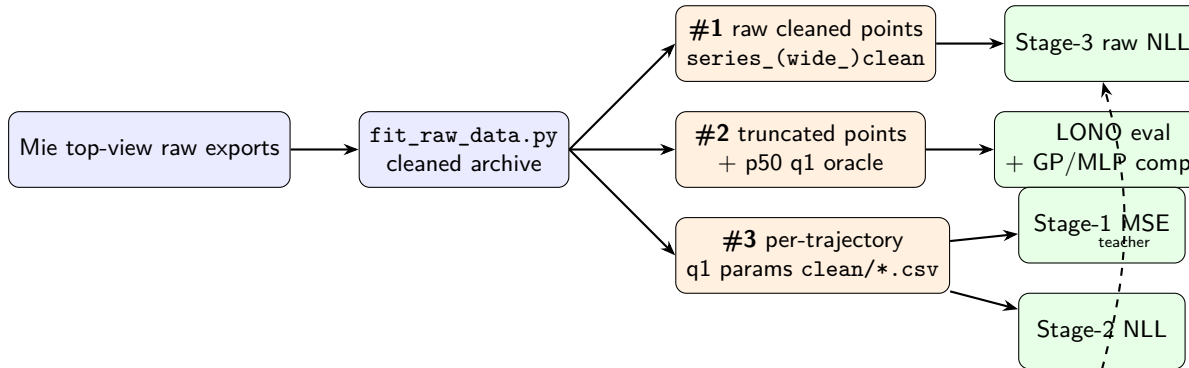
Claim: the cleaned penetration archive produced by `fit_raw_data.py` is read by the downstream pipeline through **three structurally distinct data populations**, each with its own construction protocol, selection bias, and pipeline role. They are not interchangeable, and the current thesis section 4 conflates two of them.

What this deck does:

- name the three populations explicitly and tie each one to its on-disk artefact;
- map each population to the exact training/evaluation entry point that consumes it (code-verified, not narrative);
- flag the one place where the thesis narrative is currently ambiguous (Stage-1 “representative trajectory” vs. p50 oracle);
- record the concrete edits that section 4 needs.

Scope: this is a structural planning deck. It does *not* re-derive metrics or repeat the q1 scaling story — it just reorganises what already exists so the thesis can cite it cleanly.

Where the Populations Live in the Pipeline



Three populations branch from the same cleaned archive, then re-converge only at Stage-3, where dataset #1 supplies raw supervision and the Stage-2 teacher (trained on #3) supplies the distillation prior. Dataset #2 never enters training — it is the external evaluation ruler.

Dataset #1: Raw Cleaned Penetration (Discrete Observations)

What it is. The per-frame (t, S) observations that survive cleaning, hydraulic-delay alignment, jump masks, and basic fit-quality gates. Two on-disk shapes:

- long form: MLP/synthetic_data/<nozzle>/cdf/series_clean/*.csv
- wide form: MLP/synthetic_data/<nozzle>/cdf/series_wide_clean/*.csv

Size (clean archive). $\sim 24,316$ accepted trajectories, $\sim 703,592$ individual frame observations.

Strengths.

- highest pointwise fidelity — the closest thing to a measurement record;
- preserves the full per-frame heteroscedastic noise.

Limitations.

- **right-censored** ($\sim 40\%$ of trajectories hit the FOV cap, per the audit in §4.1);
- non-uniform time sampling per condition — late bins have few survivors.

Code entry point. `load_source_table("cdf", split="clean")` in `train_stage3_distillation_plus_raw_series.py`, lines 241/247.

Used by. Stage-3 only — as **direct raw supervision** in the `raw_reliable` regime.

Dataset #2: Statistically Cleaned Points + p50 q1 Oracle

What it is. Two artefacts derived by aggregating #1 across trajectories within each operating condition:

- ① **uncensored point set** — per-condition binning at $\Delta t = 0.1$ ms, then truncation at the first bin satisfying either: smoothed trajectory count below a fixed fraction of its within-condition peak (density-drop), or per-bin p_{95} penetration reaching 95% of the FOV cap (FOV-saturation);
- ② **p50 q1 oracle** — per condition, take p_{50} of penetration in each surviving bin, fit q1 to the resulting ~ 5 –45 point trace.

Size. 506,634 uncensored points across 592 conditions; p50 oracle covers 495 conditions with median fit RMSE 0.85 mm.

Strengths.

- independent of the robust-z trajectory filter used by Stage-2 $\Rightarrow$ does not inherit the optical-quality + monotonicity selection bias of #3;
- per-condition p50 absorbs plume-to-plume noise, giving a clean ceiling for what any condition-conditioned learner could achieve.

Limitations.

Dataset #3: Per-Trajectory q1 Parameter Set

What it is. For each accepted plume trajectory, the three q1 parameters ($\log k_{1/4}$, $\log t_0$, $\log s$) fitted by Huber-NLLS to that trajectory's own (t, S) observations. One CSV per cleaned dataset under MLP/synthetic_data/<nozzle>/cdf/clean/*.csv.

Size. 24,316 accepted trajectories (clean-fit archive).

Strengths.

- smooth, differentiable, extrapolable to 5 ms via the $t^{1/4}$ prior;
- preserves between-trajectory variance required for the Stage-2 heteroscedastic NLL;
- compresses each long trajectory to three numbers $\Rightarrow$ trainable with a small MLP on a master's compute budget.

Limitations.

- inherits the robust-z selection bias (optical quality + monotonicity);
- parameters are soft-coupled (median $\text{corr}(\log k_{1/4}, \log s) \approx 0.76$) — the three components are not independent physical handles.

Code entry. `iter_clean_csv_files + select_representative_row (Stage-1) /`
`select_filtered_rows (Stage-2) in engineered_feature_common.py` lines 480/503/520

The Mapping Table (Code-Verified)

Pipeline role	#1 raw points	#2 truncated + p50 oracle	#3 q1 params
On-disk artefact	<code>series_(wide_)clean</code>	<code>cdf_points_uncensored.csv + p50_q1_fit_metrics.csv</code>	<code>clean/*.csv</code>
Stage-1 training source	—	—	● 599 representative trajectories
Stage-2 training source	—	—	● 49,908 filtered trajectories
Stage-3 raw supervision	● reliable-coverage bins	—	via Stage-2 teacher (indirect)
Stage-3 teacher distillation	—	—	● through Stage-2
In-domain test	—	—	● on #3 test split

Common Pitfall: Stage-1 “Representative” $\neq$ p50 Oracle

The two artefacts both “approximate the condition median trajectory”, so the thesis is currently easy to misread. They are not the same construction.

	Stage-1 representative	p50 q1 oracle
Source population	dataset #3 (per-trajectory q1)	dataset #2 (truncated points)
What is the “trajectory”?	one real fitted trajectory in the group	a virtual trajectory built from p_{50} per 0.1 ms bin
Selection rule	$\arg \min_i \hat{S}_i(5 \text{ ms}) - \text{median}_j \hat{S}_j(5 \text{ ms}) $	per-bin p_{50} , then re-fit q1
Inherits robust-z bias?	yes	no
Code entry	<code>select_representative_row</code>	<code>workflows/p50_q1_oracle.py</code>
Used as	Stage-1 training target	evaluation reference / oracle ceiling

Why the distinction matters for section 4. The current 4.4 text says “the group-median penetration is computed, and the nearest row is kept as the Stage-1 representative”. A reader who has read 4.3.1 (which already uses the truncated point set) can plausibly assume the Stage-1 target is the same

Cross-Population Validation Logic (the Thesis Should Highlight This)

The three populations are not just plumbing — their independence is what makes several thesis claims credible. Three places benefit from an explicit cross-reference:

(a) Scaling exponents

The $k_{1/4}$ regression on #3 gives $(a, b, c) = (0.52, -0.23, 0.26)$ with $R^2 = 0.27$. The bin-wise log-log regression on #2 gives $a \in [0.47, 0.52]$, $b \in [-0.25, -0.23]$ in the central window. **Two structurally independent populations converge on the same $\Delta P^{0.5} \rho_a^{-0.25}$ exponents** — so the empirical scaling is not an artefact of the parametric form.

(b) Selection-bias rebuttal

Stage-2's in-domain test MAE (5.63 mm) is computed on the same #3 split it trains on, and inherits the robust-z bias. The LONO MAE on #2 (4.27 mm for the production MLP per the SVGP eval deck) is bias-independent and *better* — which directly answers the “clean-plume teacher” critique that 4.2.5 raises against itself.

(c) Stage-3 supervision split

Concrete Thesis Update Plan for Section 4

New subsection (insert at end of §4.2, after the q1 diagnostics):

§4.2.6 Three derived populations and their roles in the surrogate pipeline.

One paragraph per population + the mapping table from slide 6. Explicitly state which population enters which stage.

Edits to existing subsections:

- §4.2.4 “Empirical scaling check”: add one sentence cross-referencing §4.3.1’s point-level regression as *independent* evidence on a different population. (Slide 8a.)
- §4.3.1 “Point-level scaling verification”: retitle as *Cross-population verification on dataset #2* and link back to §4.2.6.
- §4.4 Stage-1 representative description: add one sentence “This is a single real trajectory drawn from dataset #3, not the p50 oracle introduced in §4.2.6.”
- §4.2.5 selection-bias discussion: append the LONO-on-#2 evidence from slide 8b as a direct rebuttal.
- §4.5 Stage-3 regime introduction: re-phrase “raw vs. teacher” as “#1 raw observations vs. Stage-2 teacher (trained on #3)”. (Slide 8c.)

Summary

- **Three populations, one archive.** #1 raw points, #2 truncated + p50 oracle, #3 per-trajectory q1 params.
- **Distinct epistemic roles.** #1 is the pointwise measurement record, #2 is the bias-independent evaluation ruler, #3 is the compressible training target with a physical extrapolation prior.
- **Code-verified disjoint usage.** Stage-1/2 read #3 only; Stage-3 reads #1 directly and #3 through the teacher; LONO evaluation and the GP/MLP comparison read #2 only.
- **One ambiguity to fix.** Stage-1 “representative trajectory” is a real q1-fitted trajectory from #3, not the p50 oracle of #2. The current thesis text leaves this implicit.
- **Editorial, not algorithmic.** The proposed update reorganises how the thesis references existing artefacts; no new code, no new fits, no new metrics.

Hand-off for the next conversation: read slides 6 (mapping table), 7 (representative-vs-oracle), 8 (cross-population validation logic), and 9 (edit plan); apply the edits in slide 9 to Thesis/latex/sections_en/04_trajectory_surrogate_screening.tex.